

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Software and AI Engineer with **5+ years of experience** designing, developing, and delivering enterprise-scale software, data platforms, cloud-native applications, Generative AI solutions, and distributed systems using **Python, Palantir Foundry, AIP, AWS, REST APIs, microservices, and SQL**.
- Hands-on experience with **Palantir Foundry and AIP**, developing data-driven enterprise applications, AI-enabled workflows, operational solutions, ontology-backed applications, data transformations, reusable functions, and GenAI capabilities for complex business use cases.
- Experienced in building and deploying **Generative AI and LLM-powered enterprise solutions**, including agentic workflows, Retrieval-Augmented Generation (RAG), prompt engineering, embeddings, vector databases, LLM APIs, tool-use patterns, and human-in-the-loop controls.
- Strong experience embedding with business and technical stakeholders to identify high-value use cases, translate ambiguous requirements into technical solutions, rapidly prototype capabilities, and deliver production-ready software with measurable business impact.
- Proficient in **Python software engineering**, including backend services, RESTful APIs, microservices, data-processing applications, automation services, reusable libraries, asynchronous workflows, object-oriented programming, and scalable application architecture.
- Experienced with **Palantir AIP** for integrating Large Language Models with enterprise data, APIs, operational workflows, and business applications while implementing secure tool interactions, contextual grounding, and controlled human decision points.
- Strong data engineering experience involving **Palantir Foundry, Python, SQL, ETL/ELT pipelines, data modeling, data transformations, data lineage, CDC, batch processing, streaming concepts, Spark, and enterprise data integration**.
- Experienced designing **agentic AI platforms** that orchestrate LLM reasoning, enterprise APIs, structured and unstructured data, business rules, external tools, retrieval systems, and reusable AI functions.
- Hands-on cloud engineering experience with **AWS, AWS Lambda, Terraform, Infrastructure as Code, IAM, storage, compute, networking, serverless architectures, and cloud-native deployment patterns**.
- Experienced with **MLOps and LLMOps** practices including model deployment, prompt management, evaluation frameworks, model monitoring, model versioning, observability, production logging, AI workflow evaluation, and continuous delivery.
- Strong enterprise integration experience with **REST APIs, microservices, OAuth2/OIDC, JWT, SAML, event-driven architectures, secure API integrations, and distributed systems**.
- Experienced making architecture decisions that balance **quality, safety, latency, scalability, cost, maintainability, security, privacy, and model risk** for enterprise AI and cloud-native applications.
- Proven ability to independently lead technical workstreams, facilitate solution-design discussions, estimate implementation scope, mentor team members, collaborate with distributed engineering teams, and deliver software in fast-paced Agile environments.
- Strong production engineering practices including **CI/CD, Git, Azure DevOps, Jenkins, TeamCity, Docker, Terraform, logging, monitoring, versioning, documentation, runbooks, and reusable reference implementations**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming Languages:** Python, C#, Java, TypeScript, JavaScript, SQL, Bash, HTML5, CSS3.
- **Palantir Platforms:** Palantir Foundry, Palantir AIP, Foundry Data Integration, Ontology, Object Types, Actions, Functions, Workshop, Pipeline Builder, Code Repositories, Foundry APIs, AIP Logic, AIP Agents, Operational Applications.
- **Generative AI / LLM:** Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Prompt Management, Embeddings, Vector Databases, Semantic Search, LLM APIs, Context Engineering, Structured Outputs.
- **Agentic AI:** AI Agents, Agentic Workflows, Tool Use, Function Calling, Multi-Step Reasoning Workflows, AI Workflow Orchestration, Enterprise Tool Integration, Human-in-the-Loop Controls, Guardrails, Agent Evaluation.
- **AI/ML Engineering:** AI/ML Deployment, Model Integration, Model Evaluation, Feature Engineering, Experimentation, Evaluation Frameworks, Model Monitoring, Model Versioning, Responsible AI, Model Risk Management.
- **MLOps / LLMOps:** Model Deployment, Prompt Versioning, Prompt Management, Evaluation Pipelines, LLM Observability, Model Monitoring, Logging, AI Workflow Evaluation, Deployment Automation, CI/CD for AI Applications.
- **Python Engineering:** FastAPI, Flask, RESTful Services, Object-Oriented Programming, Async Programming, Data Processing, Python Libraries, Backend Development, Service Integration, Modular Architecture, Design Patterns.
- **Data Engineering:** Palantir Foundry, PySpark, Spark, SQL, ETL/ELT, Data Pipelines, Data Transformation, Data Modeling, Change Data Capture (CDC), Data Lineage, Batch Processing, Streaming Concepts, Airflow, dbt.
- **API & Microservices:** REST APIs, RESTful Web Services, Microservices Architecture, FastAPI, JSON, XML, Swagger/OpenAPI, API Gateway Concepts, Enterprise API Integration, Event-Driven Architecture.
- **Cloud Platforms:** AWS, AWS Lambda, EC2 Concepts, S3, IAM, CloudWatch, Serverless Computing, Cloud Storage, Cloud Compute, Networking, Cloud-Native Applications.
- **Infrastructure as Code:** Terraform, Infrastructure as Code (IaC), Environment Provisioning, Cloud Configuration, Infrastructure Automation, Deployment Automation.
- **Security & Identity:** OAuth 2.0, OpenID Connect (OIDC), JWT, SAML, IAM, Authentication, Authorization, RBAC, Secure APIs, Data Privacy, Enterprise Security.
- **Databases:** SQL Server, PostgreSQL, MySQL, Oracle, Vertica, Relational Databases, SQL Optimization, Data Modeling.
- **DevOps / CI/CD:** Azure DevOps, Jenkins, TeamCity, GitLab CI/CD, GitHub Actions, Git, Docker, Maven, NPM, Continuous Integration, Continuous Deployment.
- **Observability:** Splunk, CloudWatch, Application Logging, API Logging, Distributed Application Monitoring, Metrics, Production Monitoring, Root Cause Analysis.
- **Software Engineering:** System Design, Object-Oriented Design, Distributed Systems, Design Patterns, Clean Architecture, Reusable Components, API Design, Code Reviews, Documentation, Agile/Scrum.

EXPERIENCE

Software Engineer AI

Oct 2024 – Present

PCMI LLC (Full Time)

Dallas, TX, USA

- Design and deliver enterprise AI and data solutions using **Palantir Foundry, Palantir AIP, Python, SQL, REST APIs, and cloud services**, translating complex insurance, financial, and operational business requirements into scalable production applications.
- Embed with product owners, business stakeholders, architects, and engineering teams to identify high-value business opportunities, conduct technical discovery, define solution architectures, and rapidly deliver working capabilities through iterative development.
- Develop **Palantir Foundry data pipelines** that ingest, transform, reconcile, and model enterprise datasets from transactional databases, APIs, operational platforms, and downstream reporting systems into governed data products.

- Build ontology-driven applications using **Palantir Foundry Ontology**, modeling business entities, relationships, operational states, actions, and workflows to provide consistent semantic representations of enterprise data.
- Develop **Palantir AIP-powered GenAI solutions** that connect LLMs with governed enterprise data, APIs, Ontology objects, business logic, and operational workflows to automate complex knowledge-intensive processes.
- Design and implement **agentic AI workflows** incorporating LLM reasoning, tool usage, API calls, data retrieval, structured outputs, business rules, and human-in-the-loop approval mechanisms for enterprise use cases.
- Implement Retrieval-Augmented Generation solutions using **LLMs, embeddings, vector databases, semantic retrieval, prompt engineering, and enterprise knowledge sources**, improving contextual relevance and grounding of generated responses.
- Develop reusable **Python functions, Foundry transformations, APIs, microservices, and shared engineering components** that support AI applications, operational workflows, financial processing, reporting, and enterprise data products.
- Design secure enterprise integrations using **REST APIs, OAuth2/OIDC, JWT, SAML, authentication, authorization, and role-based access controls**, enabling controlled communication between Palantir applications and external enterprise services.
- Engineer complex data-processing workflows involving **CDC, transactional data, reconciliation logic, historical datasets, source-to-target transformations, batch processes, and downstream operational outputs**.
- Apply LLMOps and production engineering practices including **prompt versioning, evaluation frameworks, model monitoring, logging, observability, model-output analysis, deployment controls, and production performance tracking**.
- Make architecture decisions balancing **model quality, safety, latency, scalability, infrastructure cost, maintainability, security, privacy, and model risk** while transitioning AI prototypes into production-ready enterprise capabilities.
- Contribute reusable engineering assets including **Python libraries, Foundry transformations, prompt libraries, AI functions, API clients, reusable workflows, technical documentation, runbooks, and reference implementations**.
- Lead technical workstreams through requirements discovery, architecture discussions, sprint sizing, implementation planning, stakeholder demonstrations, production rollout, and post-deployment optimization while collaborating with distributed engineering teams.

Environments: Palantir Foundry, Palantir AIP, Foundry Ontology, AIP Agents, AIP Logic, Workshop, Pipeline Builder, Python, PySpark, SQL, REST APIs, Microservices, Generative AI, LLMs, RAG, Embeddings, Vector Databases, Agentic AI, AWS, Terraform, OAuth2/OIDC, JWT, SAML, CDC, Data Pipelines, CI/CD, Git, Azure DevOps, Agile/Scrum.

Software Engineer AI

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Designed and developed scalable healthcare applications and backend services using **Python, FastAPI, REST APIs, AWS, SQL, and microservices architecture**, supporting high-volume enterprise healthcare workflows.
- Developed Python-based RESTful microservices for processing healthcare transactions, business rules, data transformations, and integrations between internal platforms and downstream enterprise services.
- Built cloud-native and serverless processing components using **Python and AWS Lambda**, enabling event-driven workflows, asynchronous processing, API integrations, and scalable execution of backend business services.
- Developed Generative AI integrations using **Python, LLM APIs, prompt engineering, embeddings, and vector retrieval**, connecting AI capabilities with enterprise healthcare information and backend services.
- Designed Retrieval-Augmented Generation workflows that combined enterprise knowledge sources with semantic retrieval and LLM generation to provide context-aware responses for business applications.
- Developed secure REST API integrations using **OAuth2/OIDC, JWT, authentication, authorization, and role-based access controls**, supporting protected communication across healthcare applications and external services.
- Built reusable Python libraries for API communication, request processing, data transformation, exception handling, authentication, configuration management, and shared business functionality across microservices.
- Designed relational data-access components using **Python and SQL**, optimizing queries, transforming application data, implementing business rules, and supporting consistent data exchange between application and persistence layers.
- Integrated microservices with AWS storage, compute, IAM, monitoring, and serverless capabilities while applying cloud-native architecture principles for reliability, scalability, and maintainability.

- Implemented application logging, exception management, distributed diagnostics, and observability using **Splunk and AWS monitoring services**, improving visibility into service execution and production behavior.
- Integrated Python applications into **Jenkins and GitLab CI/CD pipelines**, automating builds, dependency management, packaging, version control, deployment, configuration, and environment promotion.
- Collaborated with architects, product owners, backend engineers, cloud teams, and business stakeholders during solution design, API design, sprint planning, backlog refinement, implementation, and production releases.

Environments: Python, FastAPI, Flask, AWS, AWS Lambda, REST APIs, Microservices, SQL, MySQL, Generative AI, LLM APIs, RAG, Embeddings, Vector Databases, OAuth2/OIDC, JWT, AWS IAM, Splunk, Jenkins, GitLab CI/CD, Git, Swagger/OpenAPI, Docker, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and developed enterprise backend applications using **Python, REST APIs, SQL, microservices, and object-oriented programming**, supporting scalable business applications and distributed enterprise workflows.
- Developed Python-based microservices that implemented business logic, data transformations, validation rules, API integrations, and transactional processing for enterprise application platforms.
- Built RESTful services using **Python and Flask/FastAPI** for communication between user-facing applications, backend platforms, databases, and external enterprise systems.
- Designed reusable application components using object-oriented programming, modular architecture, design patterns, configuration-driven development, exception handling, and shared Python libraries.
- Developed SQL-driven data services for processing transactional datasets, performing complex joins and aggregations, transforming business data, and exposing processed information through application APIs.
- Implemented asynchronous and batch-processing workflows in Python to handle large datasets, scheduled business processes, background operations, and high-volume enterprise transactions.
- Integrated Python applications with relational databases including **PostgreSQL, MySQL, and Oracle**, implementing data-access layers, query optimization, transactional processing, and schema-driven business logic.
- Worked with API-driven and event-oriented architecture patterns to enable loose coupling between enterprise applications, backend services, business workflows, and external integrations.
- Developed reusable API clients for JSON and REST-based services, incorporating authentication, request management, exception handling, retry mechanisms, response transformation, and application-level logging.
- Implemented CI/CD workflows using **Jenkins, Git, and Maven-based build environments**, supporting automated packaging, code versioning, deployment processes, environment configuration, and release management.
- Collaborated with product managers, architects, software engineers, and business stakeholders to convert functional requirements into technical designs, estimate development effort, and deliver enhancements through Agile sprints.
- Created technical documentation, API specifications, implementation notes, reusable development components, deployment procedures, and operational runbooks to improve maintainability and knowledge transfer.

Environments: Python, FastAPI, Flask, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Oracle, JSON, API Integration, Distributed Systems, Data Processing, Jenkins, Git, CI/CD, Docker Concepts, Linux, Agile/Scrum.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed backend application components and enterprise services using **Python, REST APIs, SQL, and microservices architecture**, supporting business-critical applications in distributed client environments.
- Designed Python modules implementing business rules, data transformations, application workflows, reusable services, configuration management, exception handling, and backend processing capabilities.
- Developed RESTful APIs using Python frameworks to expose application functionality and enable communication between web applications, mobile applications, backend services, and external enterprise platforms.
- Created reusable Python libraries and shared modules for API communication, logging, configuration, database access, data transformation, request processing, and common application functionality.
- Developed data-access layers using **Python and SQL**, supporting transactional operations, relational queries, data transformation, persistence logic, and application reporting requirements.
- Integrated enterprise applications with third-party and internal services using REST APIs, JSON payloads, authentication mechanisms, service contracts, and structured error-handling patterns.
- Contributed to microservices-based architectures by developing independently deployable backend components and defining clear service boundaries, API contracts, and integration patterns.

- Implemented batch-processing and scheduled Python workflows for large-volume data processing, automated business operations, data aggregation, and downstream system integration.
- Worked with multi-tier enterprise architectures spanning web clients, mobile applications, backend APIs, business-service layers, relational databases, and external system integrations.
- Used **Git and CI/CD practices** to manage source code, feature branches, code reviews, application builds, release artifacts, deployment workflows, and collaborative software development.
- Collaborated with senior engineers, architects, business analysts, and product stakeholders during requirement analysis, solution design, development planning, implementation, release preparation, and production support.
- Created application documentation, API specifications, development guidelines, configuration instructions, operational procedures, and reusable implementation patterns for distributed development teams.

Environments: Python, Flask, FastAPI, REST APIs, Microservices, SQL, PostgreSQL, MySQL, JSON, Backend Development, API Integration, Distributed Systems, Git, CI/CD, Linux, Cloud Concepts, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.